

DEEP ASSURANCE v6

ERC-20 Rehearsal Audit

BasicERC20RehearsalV1

PASS

completionStatus: COMPLETE

Exact source: CurveYield/Contracts @ 248e5d5de42f8a111050fb8d2b7d587653833331

Environment: Ethereum fork - block 20,000,000

Date: August 6, 2026

Prepared for CurveYield

Point-in-time AI-generated security assessment. No OpenZeppelin affiliation or certification.

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Publication Artifact Revision v2

Authoritative audit decisions remain frozen under report ID deep-assurance-v16-2-erc20-rehearsal-report-v1.

This publication revision changes only binary packaging/transport representation after rejected v1 Git blob identities.

Frozen Markdown v1 remains authoritative. The PDF content is rendered from that unchanged Markdown.

Important Notice and Engagement Metadata

This is a point-in-time AI-generated security assessment of a purpose-built ERC-20 rehearsal fixture. It uses an OpenZeppelin-inspired report organization but is not issued, endorsed, reviewed, or certified by OpenZeppelin. An audit improves assurance but cannot guarantee the absence of defects.

The user explicitly required a no-poll rehearsal. Accordingly, the poll/task bootstrap, scheduled liveness, and cross-chat wake/failover mechanics required by a production Deep Assurance campaign were intentionally not executed and are not represented as passed. All numbered agent roles, Agent 0, Agent 7, the Orchestrator, and the controller-owned final-report coordinator were executed serially by this ChatGPT / GPT-5.6 Thinking runtime. Separate-review boundaries were enforced procedurally through distinct immutable submissions and review records, but they do not constitute independent model or independent human review.

No dependencies were downloaded locally and no Solidity compilation occurred locally. Compilation and fork execution occurred only in the trusted GitHub Actions runner. No production deployment or public transaction broadcast was performed.

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The PDF version contains an automatically generated page-numbered table of contents.

Executive Summary

The rehearsal completed the Deep Assurance v16.2 phase sequence against BasicERC20RehearsalV1, a dependency-free fixed-supply ERC-20 fixture created specifically for this process test. The exact source was frozen at commit 248e5d5de42f8a111050fb8d2b7d587653833331. Manual scope, architecture, implementation, and economic-accounting lanes found no material security defect. GitHub-native compilation completed with solc 0.8.28 and zero compiler diagnostics. A GitHub-hosted Ethereum fork simulation at exact block 20,000,000 executed 24 of 24 lifecycle checks with zero failed steps and 262 archive-backed RPC requests with zero RPC failures.

The final validated issue inventory contains 0 Critical, 0 High, 0 Medium, 0 Low, and 2 Notes. The notes concern the conventional ERC-20 allowance replacement race and a zero-value transferFrom event-semantics edge case. Neither creates unauthorized value, changes total supply, bypasses allowance, or establishes a launch blocker for this fixture. No source remediation was supplied or required.

Overall security posture: PASS for the exact rehearsal fixture and exercised evidence.

completionStatus: COMPLETE

securityVerdict: PASS

Unresolved launch blockers: 0

The strongest properties observed were fixed immutable supply, absence of privileged control paths, absence of external calls/reentrancy edges, direct balance-conservation logic, finite allowance checks, and execution-backed confirmation of the infinite-allowance and self-transfer branches.

This PASS is not a certification of the Deep Assurance production orchestration layer because poll/task automation and true independent agent runtimes were intentionally omitted by request. It is also not a production deployment audit because the reviewed contract is an inert rehearsal fixture.

Exact Source

- Repository: CurveYield/Contracts
- Commit: 248e5d5de42f8a111050fb8d2b7d587653833331
- Project path: audit-fixtures/deep-assurance-v16-2-erc20
- In-scope file: BasicERC20Rehearsal_v1.sol

Final Delivery Packet

- Markdown report: Deep_Assurance_v16_2_ERC20_Rehearsal_Final_Report_v1.md
- PDF report: Deep_Assurance_v16_2_ERC20_Rehearsal_Final_Report_v1.pdf
- Supporting archive: Deep_Assurance_v16_2_ERC20_Rehearsal_Supporting_Files_v1.zip
- Evidence index: EVIDENCE_INDEX_v1.json
- Error/process records: GENERAL_ERRORS_v1.json, PROCESS_STOPPING_ERRORS_v1.json, NON_COMPLETED_PROCESSES_v1.json
- Release manifest: RELEASE_MANIFEST_v1.json
- External artifact digest receipt: ARTIFACT_DIGEST_RECEIPT_v1.json
- Publication manifest: PUBLICATION_MANIFEST_v1.json
- Publication fetch-back receipt: PUBLICATION_MANIFEST_FETCHBACK_RECEIPT_v1.json

Completion Status

completionStatus: COMPLETE

Security Verdict

securityVerdict: PASS

Final Artifact Validation

- FINAL_PDF_VALIDATED: pending until PDF render/text reconciliation
- PDF page count: pending
- Page-render QA result: pending
- Binary upload commit: pending
- Fetch-back hash equality: pending
- External artifact digest receipt: ARTIFACT_DIGEST_RECEIPT_v1.json
- Publication manifest fetch-back receipt: PUBLICATION_MANIFEST_FETCHBACK_RECEIPT_v1.json
- Final user delivery receipt: FINAL_USER_DELIVERY_RECEIPT_v1.json

These validation states are externally recorded after the report bytes are final; they are not back-filled into this authoritative Markdown in order to preserve the acyclic publication graph.

Frozen Finalization Sets

- reportId: deep-assurance-v16-2-erc20-rehearsal-report-v1
- finalizationSnapshotSha256: c5a86a09fcdf9ae74ebcd0ccf618b7f45ee6b756c5197b47fcd3c6ccdbe7305
- reportCompletionStatus: COMPLETE
- securityVerdict: PASS
- submissionAdmissionIds: admission-agent-1-scope-v1, admission-agent-2-architecture-v1, admission-agent-3-manual-v1, admission-agent-4-economic-v1, admission-agent-5-compile-v1, admission-agent-5-simulation-v1, admission-agent-6-adversarial-v2, admission-final-report-coordinator-v1
- independentReviewIds: review-first-pass-agent-7-v1, review-compile-agent-7-v1, review-simulation-agent-7-v1, review-findings-agent-7-v1, review-remediation-agent-7-v1, review-finalization-snapshot-agent-7-v1
- claimDispositionIds: claim-allowance-replacement-race, claim-zero-value-zero-source-transfer, claim-unchecked-recipient-overflow, claim-self-transfer-inflation, claim-reentrancy, claim-privilege-escalation, claim-supply-inflation, claim-finite-allowance-bypass, claim-failed-transfer-retains-allowance-decrement
- canonicalFindingIds: NOTE-01, NOTE-02
- severityDecisionIds: severity-NOTE-01-v1, severity-NOTE-02-v1
- evidenceDispositionIds: evidence-exact-source-v1, evidence-first-pass-analysis-v1, evidence-github-native-compile-v1, evidence-github-native-simulation-v1
- remediationDecisionIds: remediation-no-formal-findings-v1
- limitationDecisionIds: limitation-no-polls-v1, limitation-single-runtime-agent-simulation-v1, limitation-purpose-built-fixture-v1, limitation-no-production-deployment-v1, limitation-rpc-health-persistence-disabled-v1
- exactSetEqualityVerified: true
- PDF_CONTENT_BINDING result: pending external validation receipt
- publicationManifestFetchbackVerified: pending external validation receipt